

Inside CodeWarrior

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Global Developer Education

PalmSource
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Overview

- Metrowerks CodeWarrior IDE
- Tips and Techniques
- Q&A

Metrowerks CodeWarrior for Palm

- Metrowerks CodeWarrior IDE
  - Editor
  - Project Window (target settings)
  - Plug-ins (compiler, linker, debugger, Rez)
- Constructor (resource editor)
- Palm OS® Emulator
- Documentation
- Palm SDKs
  - Palm OS, HandSpring, Palm™ VII, Symbol

Tips and Techniques

- Documentation
- Editing
- Searching
- Constructor
- Debugging
- Build Targets
- Using POSE
- Project Stationery

Documentation Tips

- Palm Documentation
  - Get latest docs from www.palmos.com
  - Palm OS Companion
  - Development Tools Guide
  - Palm OS Reference
- CodeWarrior Documentation
  - Targeting Palm OS
  - IDE User Guide

Editing Tips

- Bring source up to date before adding code to a new file
  - Permits Header popup menu to work
- Activate browser
  - Right-click to see definition
  - Find resource constants
- Disassemble to look for really difficult problems
- Control-click to sort functions in Functions popup

Searching Tips

- Create file sets
  - Provides context for API functions
  - “I know I’ve done this before…”
- Candidate sets:
  - Palm OS 3.5 Examples
  - Palm OS 3.5 Headers
  - Palm OS Source Code
  - Project Directories

Constructor Tips

- Get the latest version
  - OS 3.5 SDK Update 1
- Build resource templates
  - Common alerts and forms
  - Difficult-to-create resources (repeating buttons)
- Duplicate resource µMWC 16400
- Invalid resources
  - Icon families and older icon types
  - Menus

Debugging Tips

- Source pane (Debugger Window)
  - Drag the current expression arrow to skip over statements
  - Alt-click to set temporary breakpoints
  - Breakpoints missing? Turn off optimizations
  - Mixed source and assembly
- Expressions Window
  - View variables in different formats simultaneously
  - View locals from calling functions in the stack

Debugging Tips

- Breakpoints Window
  - See all breakpoints in one window
  - Disable breakpoints without clearing
  - Set conditional breakpoints
  - Sorry, no watch points
    - But you can set with PalmDebugger `ss` command

Disassemble Your Source

- Can provide many answers unavailable from other sources
  - Use of global variables (look for A5)
  - Where are your constants stored?
  - Why does my code fail with optimization turned on?
- Takes time to learn
  - Don’t need to learn all the opcodes
  - Get Motorola 68K manual
  - Attend Advanced Debugging & Development class

Build Targets

- Create Debugging and Release targets
  - Build the debug target
  - Clone to build release target
- Panels
  - 68K Processor
  - Global Optimizations
  - PalmRez Post Linker

Target Setting—68K Processor

- Code Model
  - Small
  - Smart (see *Targeting Palm OS*)
- MacsBug Symbols
  - Debug build: on (new style)
  - Release build: off
- Generate A6 stack frames
  - Debug build: on
  - Release build: off

Target Setting—Global Optimizations

- Turn off for debug builds
  - What you see may not be what you get
  - Affects debugging—may prevent setting of breakpoint at statement
- Test after turning on for release builds!
  - Optimizations can have unintended side effects

Target Setting—PalmRez Post-Linker

- Type: appl
- Creator: register at www.palmos.com
- Backup bit: on
- Database name
  - Must be unique
  - Example: Sample-STRT
  - Do not leave blank!
  - Is not the same as your application's data database

Using POSE

- Profiling
- Scripting
- Host Control API
  - Controlled Gremlins and profiling
  - Logging, tracing, etc.
- Use the latest PalmPlugin

Project Stationery

- Personalized for each developer
- Elements of the perfect stationery project:
  - Release and debug targets
  - Testing target (Host Control)
  - Constructor templates
  - Utility static libraries
  - Engineering notes
  - Basic features (About, menus, prefs)
- Work in progress

Summary

- Read the documentation
- Browse the Knowledge Base
- Check for updates at www.palmos.com/dev/news

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Q&A


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